

Sourish Ghosh

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Professional Summary

Data Scientist with almost 2 years of experience building scalable pipelines and robust ML architectures, utilizing Generative AI and deep learning to solve real-life problems and extract actionable insights from complex datasets.

Professional Experience

Sravathi AI Technology Pvt Ltd [Bangalore]

Nov 2024 – Present

Associate Data Scientist

- Accelerated structured dataset generation for predictive modeling workflows by engineering scalable **data acquisition & preprocessing pipelines** for large-scale chemical datasets from **PPDB & EcoTox** using **Python & Pandas**.
- Improved predictive performance on complex chemical datasets by developing and evaluating **Machine Learning ensemble models** for toxicity prediction, incorporating feature engineering and statistical analysis.
- Engineered a **multimodal Transformer-based deep learning architecture** for **ADME prediction** using **PyTorch**, seamlessly integrating molecular representations with physicochemical descriptors.
- Enhanced domain-adapted **causal LLM** applications by fine-tuning **Gemma-2** on domain-specific datasets via **Hugging Face Transformers & SFT Trainer**, utilizing **Generative AI** prompt engineering techniques.
- Advanced **pharmacokinetic (PK) profiling** initiatives by extracting and analyzing structured datasets using **PostgreSQL** and developing robust predictive solutions through **ML & DL** approaches.
- Reduced execution time of the end-to-end **Molecular Property Prediction (MOPRP)** pipeline by approximately **40%** through comprehensive redesign and optimization using **Python & Pandas**.
- Accelerated chemical data interpretation for research teams by developing a **Streamlit-based application** for automated molecular structure generation and visualization through API-driven workflows.

IonCure Tech Pvt Ltd [Remote]

Jun 2024 – Jul 2024

Data Science Intern

- Built and maintained a structured relational database of approximately **500K–600K records** by developing automated **web scraping pipelines** across multiple distinct industry domains.
- Refined raw datasets into **300K–400K verified contacts** using **Pandas**, implementing rigorous data validation, duplicate removal & email verification workflows.
- Streamlined campaign execution and improved operational scalability by integrating **Python-based** outreach workflows with **Advanced Excel** for large-scale automation.

Projects

Generating Realistic Images using Generative Models

Jan 2024 – May 2024

- Developed a **Denoising Diffusion Probabilistic Model (DDPM)** for high-quality image generation using the Stanford Cars Dataset, trained on an **NVIDIA RTX 2080Ti GPU**.
- Optimized low-resolution to high-resolution image enhancement efficiency to **3–4 seconds** by implementing **Adversarial Diffusion Distillation (ADD)** integrated with **Blind Super Resolution (BSR)**.
- Enabled real-time conversion of low-resolution images into high-resolution outputs by developing a **Streamlit-based application** for interactive super-resolution generation.

Exploring Relationships Between Consumer Price Indices (CPI)

Sep 2023 – Dec 2023

- Conducted **time-series analysis** on Consumer Price Index (CPI) datasets during the COVID-19 period using **Python & RStudio** to evaluate relationships between economic indicators.
- Performed **Dickey-Fuller (DF) & Augmented Dickey-Fuller (ADF)** tests for stationarity assessment and evaluated multiple forecasting techniques including **AR, ARMA, ARIMA, & SARIMA**.

Customer Churn Analysis for a Telecom Company

Oct 2022 – Dec 2022

- Performed Exploratory Data Analysis (EDA) and feature-driven customer behavior assessment on telecom churn datasets using **Python & RStudio**. And conducted comparative performance analysis across classification algorithms for customer churn prediction, evaluating **Random Forest, SVM, MLP, & XGBoost**, with XGBoost achieving the highest accuracy.

Skills

- Languages:** Python, R, SQL
- Databases:** MySQL, PostgreSQL
- OS Used:** Windows, Linux, MacOS
- Tools:** Advance Excel, PowerPoint, MySQL Workbench
- Libraries:** NumPy, Pandas, SkLearn, Matplotlib, PyTorch
- Generative AI (GenAI):** Prompt Engineering, Supervised Fine-Tuning (SFT), HuggingFace

Education

Ramakrishna Mission Vivekananda Educational & Research Institute [Deemed University]

Aug 2022 – Jul 2024

Course: *MSc. in Big Data Analytics*

CGPA: 6.28

Chandernagore Government College [under The University of Burdwan]

Aug 2019 – Jun 2022

Course: *BSc. in Mathematics*

CGPA: 9.61

Achievements & Responsibilities

Placement Representative of RKMVERI CS-BDA Batch 2022-24 (with 100% Placement Record)

Jan 2024 – Jul 2024

Achieved the First place in Science of BSc. Degree College

Jun 2024

Qualified GATE 2024 in Data Science & Artificial Intelligence Paper

Feb 2024